

Topic 2.2 Limitations of Logical Inference: An example

A goal based logical agent for simplified 'Treasure Hunt: A Risky Business' [Find Gold avoiding Monsters]

a) Problem

1,4			
1,3			
1,2			
1,1	2,1	3,1	4,1

Environment: 4x4 grid; Probability of a monster at a cell except [1,1] is > 0 , and independent; Gold is hidden in some cells.

b) Assumptions

$A_{i,j}$ – the agent is in $[i,j]$

$\neg M_{i,j}$ – no monster is in $[i,j]$

$G_{i,j}$ – gold glitters in $[i,j]$

$S_{i,j}$ – smell of a monster is perceived in $[i,j]$

Actions: MoveToRight, MoveUp, ShoutWow, CryAhh, ...

Example of a percept: $S_{i,j} \wedge \neg M_{i,j} \wedge \neg G_{i,j}$

c) Initial KB

1. $S_{i,j} \Leftrightarrow M_{i-1,j} \vee M_{i+1,j} \vee M_{i,j-1} \vee M_{i,j+1}$
2. $(i \geq 1) \wedge (j \geq 1) \wedge (i \leq 4) \wedge (j \leq 4)$
3. $\neg M_{1,1} \wedge \neg S_{1,1} \wedge \neg G_{1,1}$
4. $A_{1,1}$

1,2	
$\neg M$ $\neg G$ $\neg S$ 1,1 A	2,1

d) Reasoning

- T1. $\neg S_{1,1}$ [Similarly, $\neg M_{1,1}, \neg G_{1,1}$] [Simplification; 3]
- T2. $S_{1,1} \Leftrightarrow M_{2,1} \vee M_{1,2}$ [1, 2, T1]
- T3. $(S_{1,1} \Rightarrow M_{2,1} \vee M_{1,2}) \wedge (M_{2,1} \vee M_{1,2} \Rightarrow S_{1,1})$ [Elimination of $\Leftrightarrow$]
- T4. $M_{2,1} \vee M_{1,2} \Rightarrow S_{1,1}$ [Simplification; T3]
- T5. $\neg S_{1,1} \Rightarrow \neg(M_{2,1} \vee M_{1,2})$ [Contrapositive; T4]
- T6. $\neg(M_{2,1} \vee M_{1,2})$ [Modus ponens; T1, T5]
- T7. $\neg M_{2,1} \wedge \neg M_{1,2}$ [DeMorgan's Law; T6]

So, we get $\neg M_{2,1}$ and $\neg M_{1,2}$ by Simplification of T7.

Useful addition to KB after reasoning:

5. $\neg M_{2,1}$ 6. $\neg M_{1,2}$

$\neg M$ 1,2	
$\neg M$ $\neg G$ $\neg S$ 1,1 A	$\neg M$ 2,1

Decision: **Safe to move to right or up**

Action taken, say: **MoveToRight**

Important Changes:

4. $A_{1,1}$; $A_{2,1}$

New percept, suppose: $S_{2,1} \wedge \neg M_{2,1} \wedge \neg G_{2,1}$

$\neg M$ 1,2	
$\neg M$ $\neg G$ $\neg S$ 1,1 A	$\neg M$ $\neg G$ S 2,1 A

Result of further reasoning [Self study]:

$S_{2,1}$
 $M_{2,2} \vee M_{3,1}$

Actions taken, say: MoveToLeft, MoveUp

New percept, suppose: $S_{1,2} \wedge \neg M_{1,2} \wedge \neg G_{1,2}$

Result of further reasoning:

$S_{1,2}$

$M_{2,2} \vee M_{1,3}$

So, we have: $M_{1,3} \vee M_{2,2} \vee M_{3,1}$

So, $M_{1,3}$ or $M_{2,2}$ or $M_{3,1}$ may hold, but the combination is uncertain!!!

1,3	?M		
$\neg M$ $\neg G$ S 1,2 A	?M		
$\neg M$ $\neg G$ $\neg S$ 1,1	$\neg M$ $\neg G$ S 2,1	?M	

What to do next?